

Simplification of the potassium ferrioxalate actinometer through carbon dioxide monitoring

Luke V. Lutkus, Aaron Krytenberg, Jayce Taylor, Avik Bhattacharjee, Bret A. Steinkamp, and Theresa M. McCormick 

Department of Chemistry, Portland State University, Portland, OR 97201, USA

Corresponding author: Theresa M. McCormick (email: t.m.mccormick@pdx.edu)

Abstract

Chemical actinometry can be used to determine photons absorbed for a photochemical reaction, which is required to calculate the quantum yield. A photochemical reaction with a known quantum yield can be used as a relative standard for the determination of an unknown quantum yield for a light-driven reaction. Herein, we have developed a simplified approach to using the popular potassium ferrioxalate actinometer. Traditionally, the photoreduction of Fe(III) to Fe(II) is monitored by following the absorbance of Fe(II) by reacting aliquots of the actinometry solution with 9,10-phenanthroline to form a red colored complex. The multiple steps for this method make it tedious and vulnerable to errors, especially inadvertent light exposure. In lieu of spectroscopic measurements of the Fe(II) concentration, the production of CO₂ was measured to determine the number of photons absorbed over time. CO₂ production was measured in two different ways: by the pressure increase in a sealed system and the volume change by trapping the CO₂. Both methods were considerably less laborious and showed agreeable results compared with the traditional spectroscopic method.

Key words: actinometry, potassium ferrioxalate, quantum yield, photochemistry, photon flux

Introduction

The development of photocatalysts has been of great interest, particularly in areas of green chemistry for organic transformations^{1–7} and solar fuel production, i.e., artificial photosynthesis.^{8–12} An important measurement of the efficiency of such photocatalysts is the quantum yield (QY), which is defined as moles of desired product per mole of photons absorbed.¹³ This measurement is vital in determining the applicability and scalability of a light-driven process.^{14,15} Typical QY values range between 0 and 1 in which a QY of 1 would indicate that every photon absorbed results in the desired product. QYs greater than 1 indicate subsequent dark reactions after photo-initiation. A common way of determining the QY of a photocatalyst is to use a chemical actinometer, with a known QY, to determine the number of absorbed photons, and take the ratio of absorbed photons to the product formed.¹⁶

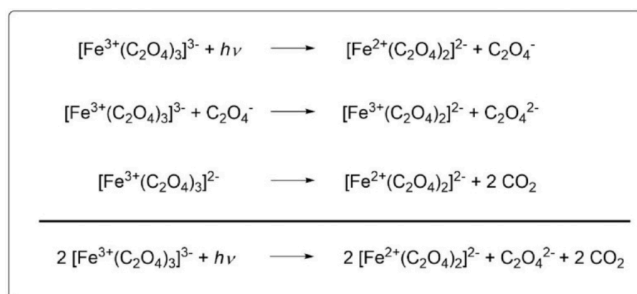
The most commonly used chemical actinometer is potassium ferrioxalate.^{17–20} It utilizes the photoreduction of potassium ferrioxalate to produce an Fe(II) species. UV-Vis spectroscopy is used to determine the Fe(II) concentration by monitoring the absorption of an Fe(II) 9,10-phenanthroline complex. Potassium ferrioxalate absorbs photons between 300 and 580 nm to produce Fe(II) oxalate and oxalate. In a simplified empirical equation representing the subsequent dark reaction, Fe(II) oxalate is oxidized by oxalate to produce Fe(III) and a dianionic oxalate species. The Fe(III) oxalate then reduces to Fe(II) oxalate by the release of carbon

dioxide. The reaction follows a 2:2:1 stoichiometry, as seen in **Scheme 1**, of Fe(II) oxalate, carbon dioxide, and oxalate per photon, respectively. This is a simplified reaction scheme that neglects association complexes and local reactions that can result in modified quantum yields with intense irradiation.²¹ The ligand 9,10-phenanthroline is added to form a visible-light-absorbing Fe(II) 9,10-phenanthroline complex. The concentration of this species is used to determine the number of photons absorbed by the sample.²²

Challenges arise with this method, since to determine the amount of Fe(II) produced, aliquots must be taken away from the light source at predetermined time points and stored in the dark for 1 h to form the visible-light-absorbing phenanthroline complex. Therefore, the rates of photons per second calculated from such measurements are often derived from only a few data points. Real-time monitoring of the potassium ferrioxalate photoreduction has been reported, where ¹³C NMR has been used to quantify the production of CO₂ rather than Fe(II).²³ The amount of photons absorbed could be calculated, but only for a limited time before the CO₂ concentration reached saturation.

Herein, we describe a simplified version to monitor the photoreduction of potassium ferrioxalate in real time by determining the amount of carbon dioxide produced by pressure change rather than spectroscopic methods. An adapted version of the pressure change method has been developed that captures the CO₂ at constant pressure; thus, changes in volume determine CO₂ production instead. Monitoring of

Scheme 1. Empirical net equation for photoreduction of potassium ferrioxalate.¹⁵ For a full reaction mechanism, see ref. 21.



CO₂ rather than Fe(II) is advantageous since multiple measurements can be obtained in real time from only one sample rather than multiple with several aliquots. Furthermore, an Fe(II) calibration curve is typically required for the traditional method but not for this method, decreasing the time required for the measurement. Here, we compare the three methods.

Experimental methods

Synthesis of potassium ferrioxalate

Potassium ferrioxalate, K₃Fe(C₂O₄)₃·3H₂O, was prepared by mixing aqueous solutions of K₂C₂O₄ (1.5 mol L⁻¹, 300 mL) with FeCl₃ (1.5 mol L⁻¹, 100 mL). Upon mixing the two solutions, an immediate change from colorless to green occurred with green crystals precipitating. The solution was left for 10 min without stirring and then vacuum filtered. The crystals were dried at 60 °C overnight.

Traditional method determining Fe(II) concentration

All solution preparation took place in a light-free setup except for a red LED light for safety. The samples were prepared by adding ~0.75 g of K₃Fe(C₂O₄)₃·3H₂O to 1 mL of 1.0 mol L⁻¹ H₂SO₄, and 9 mL of deionized H₂O to produce a 0.15 mol L⁻¹ solution in triplicate samples. The solutions were then transferred in covered containers to the light source and were irradiated with blue LEDs (Solid Apollo, 24 W 16 ft) while stirring at 500 rpm. At times 20, 40, and 60 min, a 0.1 mL aliquot was taken from each sample and added to a 100 mL volumetric flask. The aliquot was then diluted using 5 mL of an aqueous 1,10-phenanthroline solution (0.1% by weight), 1 mL of a sodium acetate buffer, and water. The solutions were shaken and allowed to remain in the dark for 1 h before the absorbance at 511 nm was measured by UV-Vis absorption spectroscopy.

Determination of carbon dioxide by pressure change

Potassium ferrioxalate solutions (0.15 mol L⁻¹, 10 mL) were prepared identically to the traditional method, covered and transported to a steel tube lined with LEDs (Solid Apollo 24 W

16 ft) that were used to evenly irradiate up to 12 samples held in a Radleys Carousel 12 cooled reaction station, and stirred at 500 rpm. The solutions were capped with caps equipped with gauge pressure sensors (Omega, PX40-100G5V, 0–15 psi, 0.15% error) and the pressure was measured every 30 s for 7 h of total irradiation time. Samples were run in triplicate.

Determination of carbon dioxide by volume change method

Solutions of potassium ferrioxalate (0.15 mol L⁻¹, 10 mL) were put in the same reaction vessel as for the pressure change and the traditional method but capped with a septum. The septum was pierced with a syringe connected to a hose fed to an inverted submerged graduated cylinder. Before irradiation, a syringe full of air was injected through the septum to evacuate any water in the hose. The solution was then irradiated with blue LEDs for 90 min and the change in volume in the graduated cylinder was recorded every 10 min. Samples were run in triplicate.

Results and discussion

Traditional spectroscopic method of Fe(II) detection

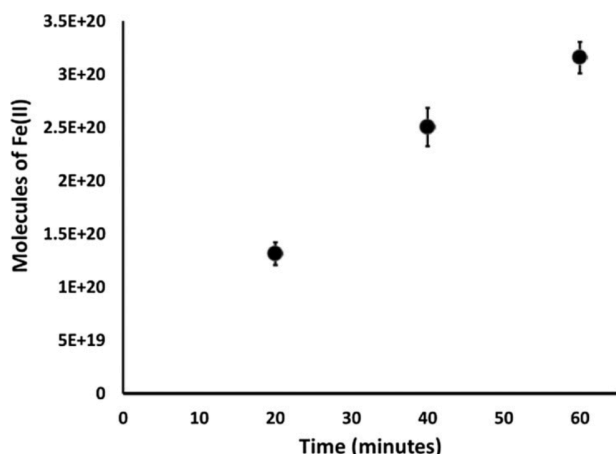
Typically, the measurement of Fe(II) produced by photoreduction utilizes the absorbance of an Fe(II) phenanthroline complex (ferroin). To quantify Fe(II) production, 11 standard calibration solutions containing ferroin are prepared. Once prepared, the solutions must be placed in the dark for 1 h before measuring the visible absorbance at 511 nm. The absorbance spectra show λ_{max} at 511 nm. The slope of the concentration vs. the absorbance at 511 nm can then be used to calculate the molar absorptivity of ferroin. An error in concentration in one calibration solution can decrease the correlation coefficient of the trend line enough that the entire calibration curve needs to be repeated.

Using the calculated molar absorptivity of ferroin, the total Fe(II) produced from the photolysis of potassium ferrioxalate can be calculated. Photoreduction of potassium ferrioxalate was performed in triplicate using this method. Solutions of potassium ferrioxalate were prepared in the dark, to avoid incident light, and then irradiated with blue LEDs. Aliquots from the solutions were removed at designated time points (20, 40, and 60 min). The aliquots were diluted and phenanthroline was added to produce ferroin. After an hour in the dark, the absorbance spectrum of each sample was taken and the amount of Fe(II) was determined using eq. 1.

$$(1) \quad n_{\text{Fe(II)}} = \frac{6.023 \times 10^{20} V_1 V_3 \log_{10}(I_0/I)}{V_2 l \epsilon}$$

where V₁ is the volume of the sample solution that was irradiated (10 mL), V₂ is the volume of sample taken for analysis (0.1 mL), V₃ is the final volume that V₂ is diluted to (100 mL), log₁₀(I₀/I) is the measured absorbance of ferroin at 511 nm, l is the path length of the spectrophotometer cell (1 cm), and ε is the molar extinction coefficient determined experimentally from the Fe(II) calibration curve (11 000 L mol⁻¹ cm⁻¹).

Fig. 1. The total Fe(II) produced over time as determined spectroscopically by absorbance of ferroin at 511 nm.



The total photons absorbed can then be calculated using the Fe(II) produced and the known QY of the photoreduction at the wavelength irradiated. Although the LEDs used are polychromatic light, the spectrum is narrow enough to treat it as monochromatic. Furthermore, the QY of potassium ferrioxalate varies only slightly for wavelengths under 480 nm. The amount of photons absorbed was calculated using eq. 2 with a QY of 0.94 for light at 480 nm.

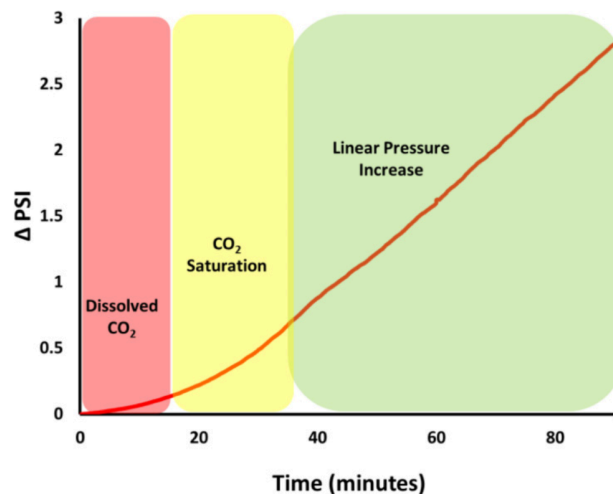
$$(2) \quad n_{\text{photons}} = \frac{n_{\text{Fe(II)}}}{\phi_{\lambda}}$$

The lengthy process results in a total of one data point per solution work-up. The process was repeated in triplicate using aliquots from three different irradiated samples. The average total amount of Fe(II) produced is shown in Fig. 1, with error bars representing one standard deviation from the mean. Care was taken to avoid incident light through the multiple steps; however, this is especially challenging when placing the cuvette in the UV-Vis spectrometer. Time points were not taken after an hour because visible precipitation of Fe(II) occurred which would result in erroneous low measurements of Fe(II) produced. Overall, this spectroscopic method of chemical actinometry results in hours of work with several possible points to introduce error.

Calculation of photons absorbed by pressure increase

Reviewing the photoreduction of potassium ferrioxalate, it can be seen that the reaction produces 2 mol of CO₂ and 1 mol of an oxalate dianion for every photon absorbed. The production of CO₂ has been previously measured in lieu of Fe(II) by ¹³C NMR; however, the reaction progress could only be monitored until the CO₂ saturated the solution and was no longer dissolved. Measuring CO₂, rather than Fe(II), has distinct advantages, including real-time measurements not requiring the formation of the phenanthroline complex. We show that the CO₂ produced may be measured by a change in pressure after saturation of the solution. The amount of

Fig. 2. Change in pressure over time from irradiation of potassium ferrioxalate (0.15 mol L⁻¹, 10 mL) with blue LEDs for first 90 min.



CO₂ produced can then be calculated using the ideal gas law in combination with Henry's law to account for the dissolved CO₂, as shown in eq. 3. However, after saturation, the slope of pressure change over time can be used to determine the rate of photons absorbed. The small change in pressure is expected to have negligible change to the equilibrium resulting in significant additional CO₂ dissolution. This rate is based on hundreds of data points rather than triplicate and does not rely on the theoretical value of Henry's law.

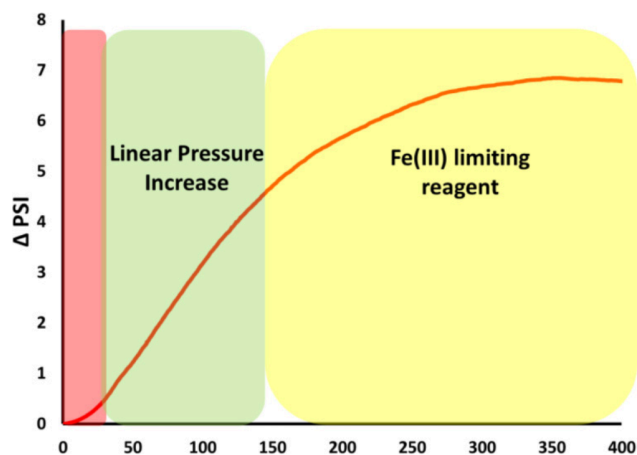
$$(3) \quad n = \frac{RT}{PV_1} + \frac{P}{K_H} \times V_2$$

where n is the amount of moles of CO₂ produced at a given time point, R is the ideal gas law constant (0.08206 L atm mol⁻¹ K⁻¹), T is the temperature in kelvin (295 K), which was held constant throughout the experiment, P is the pressure at a given time point, V_1 is the total volume (0.040 L) of the headspace of the reaction vessel, which was held constant throughout the experiment, K_H is the Henry's law constant for CO₂ at 295 K (29.41 atm mol⁻¹), and V_2 is the volume of the solution irradiated (0.01 L), which is constant throughout the experiment.

The methods using either the amount of CO₂ produced measured by pressure change or the amount of Fe(II) measured spectroscopically were compared. The conditions were kept identical using the same light source, reaction vessel, and solution volume. However, to determine the change in pressure over time, the solutions were sealed with caps equipped with pressure sensors and the pressure was measured every 30 s over the course of 7 h; the first 90 min are shown in Fig. 2.

Upon initial irradiation, the pressure does not increase due to the high solubility of CO₂. The Henry's law constant of CO₂ is 29.41 atm mol⁻¹. During this phase, the rate of CO₂ in solution cannot be calculated because Henry's law can only be used to determine the maximum amount of gas that

Fig. 3. Change in pressure over time from the photoreduction of potassium ferrioxalate (0.15 mol L^{-1} , 10 mL) with blue LED irradiation over the course of 7 h.



can be dissolved. Therefore, during the initial time period, where there is an inconsistent pressure change, the produced CO_2 remains largely dissolved in solution. Around 20 min, the pressure increase is a result of CO_2 saturation, which appears nonlinear; however, after 30 min, a linear rate of CO_2 is observed. The linear or pseudo-zeroth-order rate of CO_2 is expected as the reaction is only limited by the amount of photons absorbed. Similarly, in the traditional spectroscopic method, a linear increase in Fe(II) concentration over time is observed.

The reaction time was extended from 1 to 7 h; the resulting pressure increase with irradiation is shown in Fig. 3. This resulted in a longer dynamic range of linear CO_2 production, which can be used for calculating the amount of photons absorbed using the rate of CO_2 production. However, after 150 min of irradiation, there is a significant decrease in the production of CO_2 , which is observed as a slower increase in pressure. This region is attributed to the consumption of potassium ferrioxalate, which results in a slower rate of photoreduction. In this region, the absorption of light is no longer limiting the reaction. Measurements of CO_2 produced from these times would result in erroneously low calculations of photons absorbed. The linear region of pressure increase, however, can be used to calculate the rate of photons absorbed. The determination of the amount of photons absorbed at a given time point may be calculated using eq. 4.

$$(4) \quad n_{\text{photons}} = \frac{\text{moles of } \text{CO}_2}{\phi_{\lambda}}$$

where n_{photons} is the number of photons absorbed in moles, ϕ_{λ} is the QY of the reaction at the specific wavelength, in this case 0.94, and moles of CO_2 are determined using eq. 3 at a specific time point.

The rate of CO_2 produced can be used in a similar fashion to determine the number of photons absorbed per second or the photon flux. Utilizing the rate of CO_2 produced over time is advantageous because the slope of CO_2 produc-

tion over time is calculated based on hundreds of data points rather than a singular time point. However, it is important to note that the rate of CO_2 production should only be calculated within the dynamic range of CO_2 production, which in this case was found to be between 30 and 150 min; however, the dynamic range will depend on the intensity of the light source and should be determined by finding the linear region of pressure change. The rate of photons per second may then be calculated using eq. 5.

$$(5) \quad n_{\text{photons s}^{-1}} = \frac{\text{moles of } \text{CO}_2 \text{ s}^{-1}}{\phi_{\lambda}}$$

Ideally, the amount of dissolved CO_2 should be considered even when using the rate of CO_2 production in the linear region, since as the pressure increases, the amount of CO_2 that is dissolved in solution also increases. However, over the relatively small change in pressure of <5 psi, this effect is assumed to be negligible.²⁴ Alternatively, utilizing a change in volume can keep the pressure constant and the amount of CO_2 can be calculated using eq. 3.

Calculation of photons absorbed by CO_2 capture

To avoid the need for pressure sensors, an alternative method was developed in which CO_2 was captured using standard laboratory equipment. A rubber tube was used to connect the reaction vessel to a submerged graduated cylinder filled with water. Upon the production of CO_2 , the water was displaced. The volume of CO_2 can then be used to determine the number of photons.

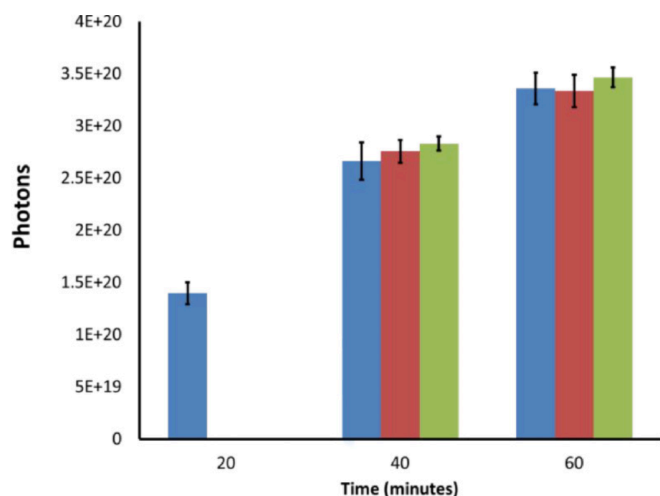
Using the change in volume to follow the photoreduction of potassium ferrioxalate, the equation to calculate the amount of CO_2 produced is the same as eq. 3, except the pressure is considered constant while volume (V_1) changes instead. Therefore, amount of CO_2 dissolved is calculated using Henry's law at a constant pressure and temperature.

The change in volume over time showed similar trends to monitoring the change in pressure over time. The initial period up to 20 min was not used because CO_2 was not yet saturated in solution. The measured amount of CO_2 at 40 and 60 min was in good agreement with the prior two methods. The results from each of the three methods (Fe(II) production, change in pressure, and change in volume) can be seen in Fig. 4.

Conclusions

Chemical actinometry is a necessary photochemical measurement to determine the QY of a light-driven reaction. The photoreduction of potassium ferrioxalate is one of the most common chemical actinometers used. The amount of solution preparation, transfers, and waiting periods make this method tedious and leave room for error. For these reasons, QY measurements typically have an assumed 10% margin of error. We have shown two alternative approaches to this method, which rely on measuring the amount of CO_2 produced by physical methods rather than the amount of Fe(II) , which is typically measured spectroscopically. Measuring CO_2

Fig. 4. Calculated number of photons absorbed by the three methods: (1) traditional method (blue), (2) pressure sensor method (red), and (3) CO₂ capture method (green).



comes with its own advantages and disadvantages compared with the traditional spectroscopic method.

In both CO₂ measurements, by volume or by pressure, at early time periods (prior to 30 min of irradiation), it was difficult to calculate the amount of CO₂ produced. This is a result of the high solubility of CO₂ in water. Further research may focus on electrolytes or other compounds to lower the solubility of the CO₂ to avoid such shortcomings or presaturating the solution with CO₂ prior to irradiation. However, it should be considered that the effect of additional chemical species in solution may alter the QY of the reaction.

The advantage of measuring CO₂ at later time periods, between 30 and 150 min, is the decreased chance for incident light exposure as well as possibly addressing concerns of inaccurate measurements at high irradiation, although more study is needed to confirm this advantage.²¹ The samples were not moved during the course of the measurement and the measurement was done in real time. Furthermore, the vast amount of data points that can be collected from a single sample is far superior to the Fe(II) detection method that yields only one data point per sample. There is no dissolution or waiting period in the dark, which is also beneficial for adopting this methodology for widespread implementation.

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Data availability

Data are available upon request.

Author information

Author ORCIDs

Theresa M. McCormick <https://orcid.org/0000-0003-1745-2911>

Author contributions

Conceptualization: LVL, TMM

Data curation: AK, JT, BAS

Formal analysis: LVL, AK

Funding acquisition: TMM

Investigation: AK

Methodology: AK, JT

Supervision: BAS, TMM

Visualization: AB

Writing – original draft: LVL, AK

Writing – review & editing: AB, BAS, TMM

Competing interests

The authors declare there are no competing interests.

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